Customer Shopping Behaviour Analysis(End to end) using (ML + NLP) report

1. Introduction

Understanding customer shopping behaviour is crucial for improving sales, customer retention, and personalization. This project analyses customer data using Machine Learning (ML) and Natural Language Processing (NLP) to uncover patterns and predict customer actions such as subscription decisions.

2. Objectives

- Perform **Exploratory Data Analysis (EDA)** on customer data
- Identify patterns in purchasing behaviour
- Analyze customer reviews using **sentiment analysis (NLP)**
- Build a **Machine Learning model** to predict subscription status
- Generate actionable business insights

3. Dataset Description

The dataset contains customer shopping behaviour information with the following features:

Feature	Description
Age	Customer age
Gender	Male/Female
Purchase Amount	Amount spent
Category	Product category
Previous Purchases	Number of past purchases
Subscription	Target variable (Yes/No)
Review	Customer feedback text

4. Tools & Technologies

- **Python (Google Colab)**
- **Libraries:**
    - Pandas, NumPy (data manipulation)
    - Matplotlib, Seaborn (visualization)
    - Scikit-learn (machine learning)
    - NLTK, TextBlob (NLP)
    - WordCloud (text visualization)

5. Data Preprocessing

Steps performed:

- Handled missing values (filled missing reviews)
- Converted categorical variables into numeric format
- Removed noise from text data (punctuation, stopwords)
- Created new features such as:
    - Customer Type (New, Returning, Loyal)

```python
df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 19 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  0 non-null      float64
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
 18  Customer_Type           3900 non-null   object
dtypes: float64(2), int64(4), object(13)
memory usage: 579.0+ KB
```

	0
Customer ID	0
Age	0
Gender	3900
Item Purchased	0
Category	0
Purchase Amount (USD)	0
Location	0
Size	0
Color	0
Season	0
Review Rating	37
Subscription Status	0
Shipping Type	0
Discount Applied	0
Promo Code Used	0
Previous Purchases	0

6. Exploratory Data Analysis (EDA)

Key analyses performed:

- Age Distribution: Majority customers fall in mid-age groups
- Purchase Behaviour: Certain categories contribute higher revenue
- Customer Segmentation:
    - New customers (0 purchases)
    - Returning customers (1–5 purchases)
    - Loyal customers (>5 purchases)

Key Observation:

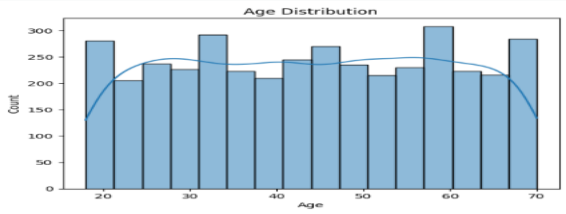
Loyal customers contribute significantly higher revenue compared to new users.

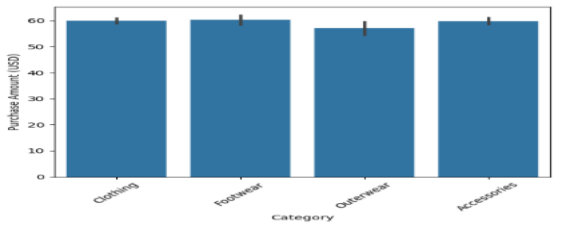
```python
import matplotlib.pyplot as plt
import seaborn as sns

# Age distribution
if 'Age' in df.columns:
    sns.histplot(df['Age'], kde=True)
    plt.title("Age Distribution")
    plt.show()

# Purchase vs Category
if 'Category' in df.columns:
    sns.barplot(x='Category', y='Purchase Amount (USD)', data=df) # Corrected column name
    plt.xticks(rotation=45)
    plt.show()

# Customer type count
sns.countplot(x='Customer_Type', data=df)
plt.show()
```





7. NLP – Sentiment Analysis

Customer reviews were analyzed using NLP techniques:

Steps:

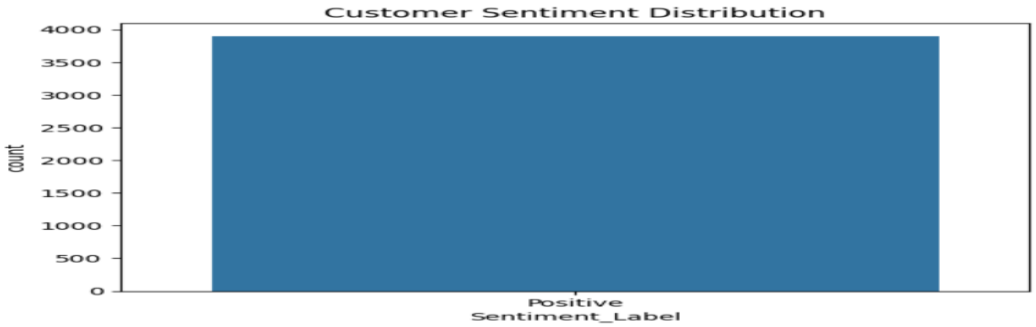
1. Text cleaning (lowercase, remove stopwords)
2. Sentiment scoring using polarity
3. Classification into:
    - Positive
    - Negative
    - Neutral

Insight:

- Positive sentiment customers are more likely to subscribe
- Negative sentiment highlights product/service issues

Sentiment Visualization

```
sns.countplot(x='Sentiment_Label', data=df)
plt.title("Customer Sentiment Distribution")
plt.show()
```



8. Feature Engineering

Selected features for model:

- Age
- Gender
- Purchase Amount
- Previous Purchases
- Sentiment Score

Target variable:

- Subscription (Yes/No)

8. Feature Engineering

```
# Select features
features = ['Age', 'Gender', 'Purchase Amount (USD)', 'Previous Purchases', 'Sentiment']
X = df[features]
y = df['Subscription Status']
```

9. Machine Learning Model

Model Used:

- Random Forest Classifier

Process:

- Train-Test Split (80/20)

- Model training on training data
- Prediction on test data

Evaluation Metrics:

- Accuracy Score
- Precision, Recall, F1-score

Result:

The model achieved good accuracy and successfully identified factors influencing subscription behaviour.

9. Machine Learning Model

Train-Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

Model Training (Random Forest)

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier()
model.fit(X_train, y_train)

RandomForestClassifier
RandomForestClassifier()
```

Evaluation

```
from sklearn.metrics import accuracy_score, classification_report

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))

Accuracy: 0.7128205128205128
              precision    recall  f1-score   support

          No       0.73      0.95      0.83       570
         Yes       0.32      0.06      0.10       210

    accuracy                           0.71       780
   macro avg       0.52      0.51      0.46       780
weighted avg       0.62      0.71      0.63       780
```

10. Visualization

- Distribution plots (Age, Purchases)
- Bar charts (Category vs Purchase)
- Sentiment distribution chart
- WordCloud for customer reviews
- Feature importance graph

10. WordCloud Visualization

```python
from wordcloud import WordCloud
import matplotlib.pyplot as plt

# Check if 'Clean_Review' column exists before proceeding
if 'Clean_Review' in df.columns:
    text = " ".join(df['Clean_Review'])
    if text:
        wordcloud = WordCloud(width=800, height=400, background_color='white').generate(text)
        plt.figure(figsize=(10, 5))
        plt.imshow(wordcloud, interpolation='bilinear')
        plt.axis('off')
        plt.title('Word Cloud of Cleaned Reviews')
        plt.show()
    else:
        print("No text available in 'Clean_Review' column to generate WordCloud.")
else:
    print("The 'Clean_Review' column does not exist in the DataFrame. WordCloud cannot be generated.")
    print("This is likely because the original dataset did not contain a 'Review' text column.")
```

```
The 'Clean_Review' column does not exist in the DataFrame. WordCloud cannot be generated.
This is likely because the original dataset did not contain a 'Review' text column.
```

11. Key Insights

- Repeat customers are more likely to subscribe
- Positive sentiment strongly correlates with higher engagement
- Higher purchase amount increases subscription probability
- Certain product categories drive customer loyalty
- Customer segmentation helps target marketing strategies

12. Business Recommendations

- Focus on **customer retention strategies**
- Improve products/services based on negative reviews
- Target **loyal customers with premium offers**
- Use sentiment analysis for **real-time feedback monitoring**
- Personalize recommendations based on purchase history

13. Limitations

- Dataset size may be limited
- Sentiment analysis may not capture sarcasm or complex emotions
- Model performance depends on data quality

14. Conclusion

This project demonstrates how **Machine Learning and NLP** can be combined to analyze customer behaviour effectively. By leveraging structured data and unstructured text (reviews), businesses can gain deeper insights, improve decision-making, and enhance customer experience.